

TABLE 1 Gravidity, gestational age and crown-rump length (CRL) in emetic and nonemetic pregnancies

	Emesis	Nonemesis	Emesis vs. nonemesis
Subjects (N)	26	17	
Gravidity*	2.7 ± 0.31	1.4 ± 0.18	P < 0.01
CL dx	2.6 ± 0.34	—	
CL sin	3.0 ± 0.68 ^{ns}	1.5 ± 0.20	P < 0.01
Gestational age (days)			
Total number	71 ± 2.56	67 ± 3.81	NS
CL dx	67 ± 2.56	—	
CL sin	79 ± 4.75 ^{0.05}	68 ± 4.19	NS
CRL (mm)	35.8 ± 3.94	30.4 ± 5.51	NS
CL dx	29.6 ± 3.81	—	
CL sin	48.1 ± 7.62 ^{0.05}	32.4 ± 6.10	NS

Only two nonemetic women had their corpus luteum on the right side and were therefore excluded from calculations of means and standard deviations.

* CL, corpus luteum; dx, right side; sin, left side; NS, not significant.

sea and vomiting. The physiological cholestasis of pregnancy combined with high estrogen levels may play a part in the syndrome of nausea and vomiting of pregnancy. It is recognized that some patients can not tolerate the oral contraceptive tablets and it may be their content of estrogen that causes these patients' nausea. It will be interesting to see what becomes of the corpus luteum theory. It is hard to imagine that

nature made this mistake. It is good news that investigators are now looking for an anatomical and physiological cause for this early symptom of pregnancy instead of fixing the blame on psychological causes.

With the ability to provide almost total nutritional requirements intravenously, nausea and vomiting of pregnancy rarely becomes a serious problem. In past generations, however, some patients died from complications of nausea and vomiting of pregnancy. The older textbooks classified vomiting of pregnancy under toxemia of pregnancy. Patients died of dehydration, starvation, and multiple vitamin deficiencies. Termination of pregnancy was often used as part of the treatment, but the procedure itself often proved fatal because of the precarious condition of the patient. Charlotte Brontë (1816–1855) died of complications of nausea and vomiting of pregnancy. It was not too long ago that patients with persistent nausea and vomiting were isolated from family, seen by a psychiatrist, and tube fed (a barbaric practice).

There are very few who now think that persistent nausea and vomiting of pregnancy arise from emotional problems in the patient's life or in her relationships with others.—Ed.)

Umbilical Artery Flow Velocity Waveforms in High-Risk Pregnancy: Randomised Controlled Trial

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Flow velocity waveforms from the umbilical artery in the cord may be recorded by Doppler ultrasound examination. The present authors report a randomized controlled trial to investigate the effect of Doppler umbilical artery flow

studies on perinatal practice. They sought to determine whether the early identification of the potentially compromised fetus by this method would result in a change in the timing of delivery and obstetrical intervention.

Three hundred patients were enrolled over 1 year, 167 being allocated to the control group and 133 to the Doppler report group. The two groups did not differ with regard to obstetrician, health insurance status, parity, maternal age, or reason for hospital admission.

The subsequent obstetrical management of the two groups is shown in Table 1. Eleven patients were delivered at other hospitals (five control, six report) and were excluded from analysis. The timing of delivery was similar for the two groups. Emergency cesarean delivery and fetal distress (repetitive late decelerations of the fetal heart rate, or fetal scalp blood pH less than 7.2) during labor were more frequent in the control group than in the report group.

The neonatal outcome was better in the report group than in the control group. In the latter, the proportion of infants requiring neonatal intensive care (level 3) admission was greater, and the mean hospital stay was longer. These differences were not significant. Infants from the control group ($N = 162$) spent a total of 560 days in the level 3 nursery (group mean stay, 3.46 days), whereas infants from the report group ($N = 127$) spent only 220 days in the level 3 nursery (group mean stay, 1.73 days). The requirement for ventilatory support in the neonatal period was greater in the control group ($P = 0.10$), and there was a similar difference in the requirements for oxygen therapy (31 control, 17 report).

There were two stillbirths in the control group, one with multiple fetal abnormalities and one unexpected intrauterine death at 37 weeks. The stillbirth in the report group was a case of fetal ascites with other anomalies. The three neonatal deaths were all in the control group. Two were associated with fetal abnormalities, namely, a case of diaphragmatic hernia with other anomalies and a case of

TABLE 1 Obstetrical management

Index*	Control ($N = 162$)	Report ($N = 127$)	P
Gestation at 1st investigation (mean, wk)	34.5	34.5	NS
Gestation at delivery (mean, wk)	37.8	38.0	NS
Spontaneous onset	37.1	38.1	NS (0.07)
Elective (CS, induction)	38.6	38.2	NS
1st investigation—delivery interval (mean, days)	23.7	24.8	NS
Period in hospital after enrollment (mean, days)	9.7	8.5	NS
Labor onset:			
Spontaneous onset labor (no)	55 (34%)	45 (35%)	NS
Induction of labor (no)	79 (49%)	57 (45%)	NS
Elective CS (no)	28 (17%)	25 (20%)	NS
Emergency (in labor) CS:			
Total cases (% of all labors)	31 (23%)	13 (13%)	<0.05
After spontaneous onset of labor (% spontaneous labors)	19 (35%)	6 (13%)	<0.025
After induction of labor (% induced labors)	12 (15%)	7 (12%)	NS
Fetal distress in labor:			
Total cases (% of all labors)	11 (8%)	3 (3%)	0.075
After spontaneous onset	8	1	0.03
After induction	3	2	NS

* CS, cesarean section; NS, not significant ($P > 0.05$).

visceral asymmetry syndrome in polysplenia. The third occurred after premature rupture of membranes and delivery at 29 weeks.

(This study demonstrates the clinical application of a technique for measuring umbilical artery waveforms through Doppler studies. Blood flow in the umbilical artery of the growth retarded infant is reduced when compared to the normally growing infant. In the hands of Trudinger and his colleagues, recognition of reduced umbilical artery flow was a more effective method for recognition of beginning fetal hypoxia than the traditional methods, such as electronic fetal heart rate monitoring, and fetal movement counting. Investigations by others have shown reduced blood flow in uterine arcuate arteries that are characteristic of intrauterine growth retardation (S. Campbell and others, *Lancet* 1: 675, 1983; *Survey* 38: 548, 1983). The placen-

tas of growth retarded infants have a poor blood supply, particularly in cases of maternal hypertensive diseases. Approximately 25 per cent of growth retarded infants are born to mothers with essential hypertension or who have pre-eclampsia or eclampsia. Studies of the placental beds and placentas of patients with hypertensive complications of pregnancy and of patients with idiopathic fetal growth retardation have shown failures of proper placentation. Studies of spiral arterioles of decidua and myometrium have revealed inadequate invasion of spiral arterioles by trophoblastic cells resulting in reduced circulatory supply to the placenta (T. Y. Khong and others, *Br. J. Obstet. Gynaecol.* 93: 1049, 1986; *Survey* 42: 503, 1987).

I do not know any groups in this country making clinical decisions concerning fetal health and time of delivery by studying umbilical artery blood flow. There are several groups studying arcuate artery blood flow during normal and abnormal pregnancy and several that are measuring fetal aortic blood flow. Investigations of this type are being correlated with fetal heart

activity, growth studies, and studies of fetal blood pH, oxygen, and carbon dioxide levels. P. W. Soothill and others of King's College, London, have correlated fetal blood gases obtained by cordesis with fetal aortic blood flow studies made with Doppler equipment (*Lancet* 2: 1118, 1986; *Survey* 42: 443, 1987).—Ed.)

Reliability of Ultrasound Fetometry in Estimating Gestational Age in the Second Trimester

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The present study was undertaken to serve three purposes: 1) to evaluate the reliability of biparietal diameter (BPD), occipitofrontal diameter (OFD), mean abdominal diameter (AD), and femur length (FL) measurements as indicators of gestational age (GA); 2) to determine whether these variables, separately or in combinations, were superior to biparietal diameter alone in estimating gestational age in women with known date of conception; and 3) to determine whether the inclusion of several fetometric variables in routine ultrasound examination would reduce significantly the number of postterm pregnancies.

The study was performed in several stages. In the first stage, the reliability of measurements of the four variables, alone or in combination, as indicators of true gestational age was evaluated. Fourteen fetuses were examined longitudinally on five occasions in the 11, 13, 16, 20, and 24 completed weeks ($N = 67$) (Fig. 1). In all 14 cases, the women had regular 28-day menstrual cycles, and a reliable date of conception was available from basal body temperature charts. A crown-rump measurement at 10 weeks confirmed the dates in all women, and none was excluded.

In stage 2, the results from stage 1 were tested. The best equations obtained were used to estimate gestational age in 44 women for whom true gestational age was known from the date of artificial insemination.

In stage 3, of the 2289 women routinely examined by ultrasound at approximately gestational week 16 (i.e., more than 97 per cent of those who gave birth at the

authors' hospital), 2021 who subsequently gave birth spontaneously were included in the study, in which biparietal diameter and femur length were used to estimate gestational age.

Table 1 gives the equations that were found to provide the highest precision in estimating gestational age in the interval 80 to 180 gestational days. Of the individual variables, biparietal diameter reflected true gestational age with a standard deviation of the error of 3.2 days, and femur length with an error of 3.9 days. When all variables were combined in a formula, gestational age could be estimated with an error of 2.2 days. No significant improvement was found by combining more variables than biparietal diameter and femur length (Fig. 1).

In the 44 women with known dates of conception, gestational age estimated by biparietal diameter differed from true gestational age, on average, by 0.009 days and had a standard deviation of 3.2 days (range, -8 to $+6$ days). Estimated gestational age based on femur length was below true gestational age by -0.57 days, and the variation was 3.4 days (range, -9 to $+5$ days). A formula combining biparietal diameter and femur length values ($GA = BPD \times 1.2 + FL \times 1.0 + 49$) gave a mean difference of -0.55 days and a standard deviation of 2.7 days (range, -7 to $+5$ days).

For the whole group giving birth spontaneously in Malmö during 1982, estimated gestational age at delivery was, on average, 0.43 days less when based on femur length than